

An Impact of YOLOv5 on Text Detection and Recognition System using TesseractOCR in Images/Video Frames

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Abstract—Text detection and recognition in images and videos are significant research areas in computer vision. A computer vision technology is used for smart city real-time traffic monitoring, and a security camera can simultaneously record the license plate information of suspected vehicles. The challenging task here is detecting the text images that are arbitrary oriented, such as aerial photographs and scene texts. Most complementary text detection and recognition methods are designed to identify text in images that are clear in the background and near-horizontal text. However, those methods will not be effective in detecting text in complex images and video streams. To address this issue, we propose a system that detects the text images using the YOLOv5s model, which effectively trains small-scale images and YOLOv5x for large-scale images. TesseractOCR recognizes the detected text by converting the image to a string and storing it in CSV format. The experiment was carried out for ICDAR2013, ICDAR2015, and YVT images/frames. The results indicate that the proposed method using YOLOv5x effectively detects images/video frames with reasonably good accuracy, and the recognition rate is suitable for a near-horizontal image using TesseractOCR.

Keywords—Deep Learning, ICDAR2013, ICDAR2015, Text Detection, Text Recognition, TesseractOCR, YOLOv5, and YVT Dataset.

I. INTRODUCTION

Text is the most important means of communication and plays an essential role, and it can be embedded in documents and natural scenes (images) as an information carrier. Text detection is a critical component of various computer vision-based applications, including Robotics, Industrial Automation, Image Search, Instant Translation, Automotive Assistance, and Sports Video Analysis [1, 2]. The text detection method is mainly used to identify the text from scanned or printed documents. Detection of text in an image is challenging because the text will be exposed to different font sizes, lighting effects, and environmental conditions [3].

In recent years, detection and recognition of text from an image/video frame is in demand because of its application in many circumstances like to read the credit card number, Aadhaar/PAN card number, name, billing amount, billing date, and other information from the paper. Optical Character Recognition (OCR)[4, 5] is a popular text recognition method that converts scanned text images into editable formats. It can be challenging to detect and recognize the words, but deep

learning allows one to read text from images effectively. This paper proposed an efficient and customized text detection and recognition in images/video frames using YOLOv5 and TesseractOCR methods.

II. RELATED WORK

The related works for text detection and recognition in images are discussed and summarized. Recently, deep-learning-based methods have significantly improved text performance of both detection and recognition. The work follows text detection using YOLOv5, which detects the text region by creating a bounding box and passing the detected bounding box region to the TesseractOCR engine for recognition.

A. Text Detection

The text has received increasing attention in videos and images as a direct and critical information source. For example, caption text frequently provides information concerning where and when the events in the video occurred and who was involved. Signpost text is commonly used as a visual indicator for scene notifications and navigation [6]. For text detection in images/video frames, YOLO-You Only Look Once is used as an object detection framework. There are multiple types of YOLO depending on their size and space YOLOv5s is the smallest yet fastest variant. YOLOv5m is a medium-sized model slower than YOLOv5s but faster than other variants. This system requires only a camera and a computer for detection [7].

The PSENet model is used for scene text detection to achieve an efficient and accurate result and is used to tackle the problem of false detection and missed detection for most scene text detection methods [8]. Localizing the individual character regions and connecting the detected characters to a text instance, CRAFT: Character-Region-Awareness-For-Text-detection is used. CRAFT predicts two scores/maps for each character: Region-Level-Maps and Affinity-Level-Maps [9]. However, the CRAFT model still needs to work on the text recognition part.

B. Text Recognition

Text Recognition is the process of recognizing text characters from an image/video frame that belongs to the area of pattern recognition. OCR (Optical Character Recognize) is a text recognition software used to extract or recognize the text

in images/video frames [10, 11]. It separates different characters from each other, taken from an image or a video. An OCR method includes two main components: Text Detection (TD) and Text Recognition (TR); text detection aims at localizing all text sections within the given image, either at Word-level/Text-Line-Level. Meanwhile, text recognition is used to extract the visual signal into natural language text. Text recognition is framed as a Coder-Decoder task in existing CNN techniques. Here, a CNN-based encoder is used for image analysis, and an RNN-based decoder is used for text creation [12].

CNN is the best approach for recognition, and its performance is better than ANN. It takes the input as an image or video and allocates the learnable weights to various aspects to discriminate from others [13]. In natural scene images, a text appears in various orientations, such as Horizontal-oriented, Curved, Arbitrarily-Oriented, and Vertically-OrientedTexts; hence recognizing text is very challenging. MTD (Multi-directional-Text-Detector) is used to detect a different text orientation in natural scene images [14]. The combination of Tesseract and deep learning will make the model highly effective for translating or converting detected text to display as output. This approach is commonly used to extract data present in an image [15]. The present recognition techniques are not robust and precise to detect the text area, and they also have trouble recognizing distorted, rotated dynamically, or out-of-order text. In conclusion, recognizing a pattern of text in unstructured data is challenging.

Based on the literature suggests that text in various orientations is still a topic for researchers.

III. PROPOSED FRAMEWORK

In this paper, the YOLOv5 model is proposed for text detection, based on a Conventional neural network (CNN) that is solely used for object detection, but challenging work is to detect the text in images.

The YOLO approach divides the input data (image) into distinct grids depending on predetermined grid sizes. Then it indicates the likelihood of the intended text at every grid; in a single run, it will predict the text boundaries, which is effective in real-time detection. And then pass the detected boundary box to text recognizer TesseractOCR [16], extracting the text from the image. Fig. 1 describes the task involved in each pipeline.

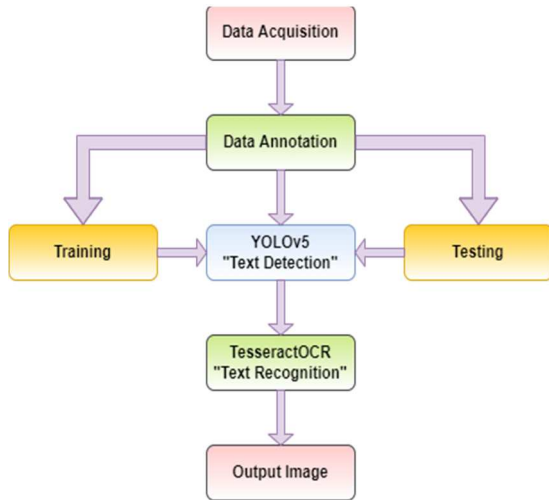


Fig. 1. Text Detection and Recognition System workflow

A. Data Acquisition

The first step is to acquire the data; for that, we used benchmark datasets ICDAR2013 [17], ICDAR2015 [18], and YVT (YouTube Video Text) [19]. The ICDAR Robust Reading Competition for 2013 is divided into three challenges focusing on text extraction in several application domains, including born-digital photos, real scene photographs, and actual scene films. This dataset contains 229 training images and 233 testing images. ICDAR2015 Robust Reading Competition is featured for scene text detection and localization. It has 1500 images, and we will be using 1000 images to train and the rest for testing; the dataset here contains small text and blurred images.

Furthermore, YVT (YouTube Video Text) dataset consists of 30 videos in the form of frames. Each movie was taken from YouTube and is 15 seconds long with 30 frames per second. The datasets are split into an 80 to 20 ratio for training and testing in the proposed work.

B. Data Annotation

The dataset should be appropriately annotated for the training and testing model, which is done by using Roboflow [20]. Roboflow is used to annotate the text in an image by putting the bounding box around the text region manually and helps us to save the annotated dataset to the desired format; in this case, it is .txt, where it is divided into two segments. 80% of the dataset is used for training, and 20% is used for testing in the model.

C. Develop Deep Neural Network using YOLOv5

In Deep learning (DL), feature extraction is done automatically, but it is manually done in machine learning. Convolutional neural networks (CNN) is the most well-known DL model for image/video frames data training. Fig. 2 shows the Deep Learning architecture of YOLOv5. YOLOv5 is used because it is rapid and compact and as well as requiring significantly less processing resources than another cutting-edge detection model. YOLOv5 is much faster compared to the previous versions of YOLO. YOLOv5 mainly consists of the backbone, neck, and final layer. YOLOv5 uses CSPNet(Cross Stage Partial Network) [21] as its backbone for feature extraction such as edges and shapes and to extract valuable characteristics from images. It also uses PANet (Path Aggregation Network) as its neck to boost information flow and is designed to perform Multi-scale object prediction. YOLO layer (1x1 Conv) is used as the head of the final layer and is a final detection step.

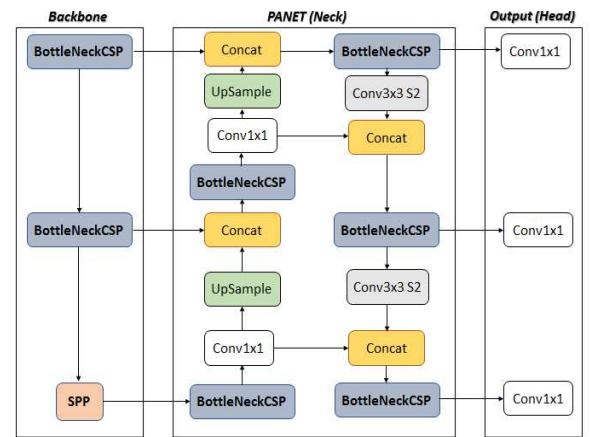


Fig. 2. YOLOv5 Architecture [22]

D. Text Recognition

Text recognition is used to recognize text in image/video frames. Optical Character Recognition (OCR) extracts the detected word from the image and achieves good results. TesseractOCR comprises sequential stages: adaptive binary thresholding, assessing the connected components and their connections, character recognition, and character aggregation to form words, as shown in Fig. 3. Text will be extracted using TesseractOCR and given to the text file by cropping the text block.

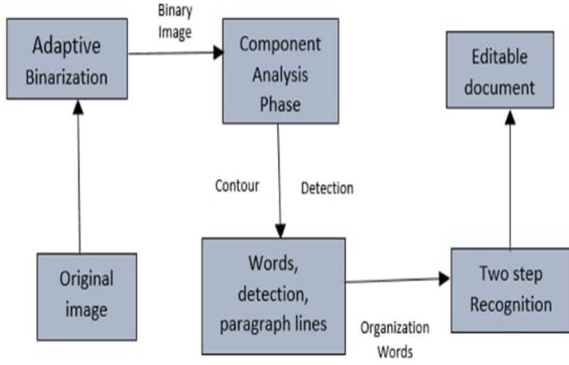


Fig. 3. TesseractOCR Architecture

IV. IMPLEMENTATION AND RESULTS

In the present work, the experiment is carried out in English text datasets. The excellent results were achieved using Python 3.6 on a 64-bit Intel Core I5 -3.2 GHz PC with an NVIDIA(R) RTX 3070 GPU. The benchmark datasets used are ICDAR2013, ICDAR2015, and YVT video datasets to evaluate the proposed method. The dataset is separated into two sections: training and testing data. Here, training was done using 80% of the data, and testing was done with the remaining 20%. The model can detect text in a frame on real-time running video with 74% accuracy. The comparison result of detected and recognized text in challenging images/video frames using YOLOv5s and YOLOv5x, as shown in Fig. 4 and Fig. 5. Here, yolov5x can detect text with very high accuracy.



Fig. 4. Results of text detection and recognition of images/ video frames using YOLOv5s

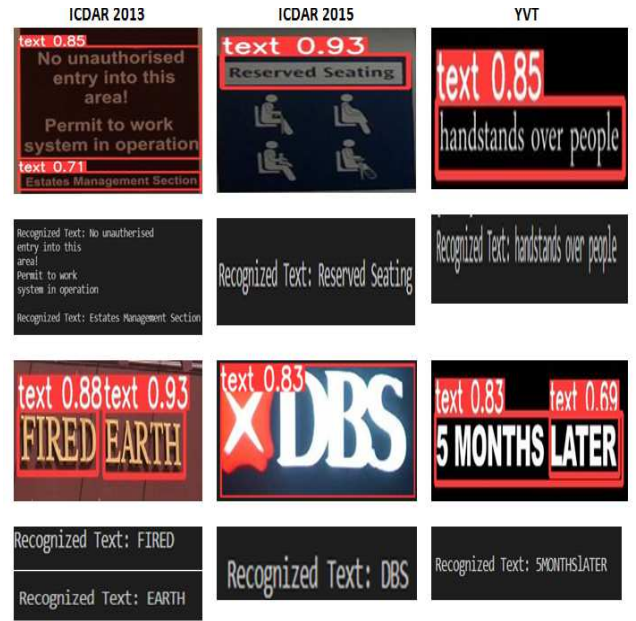
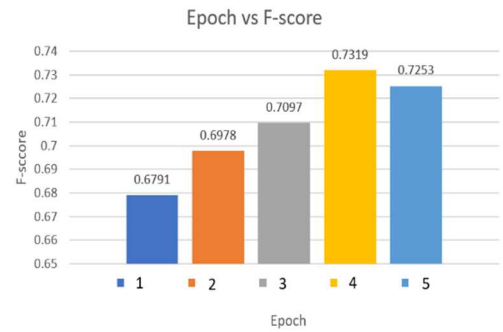
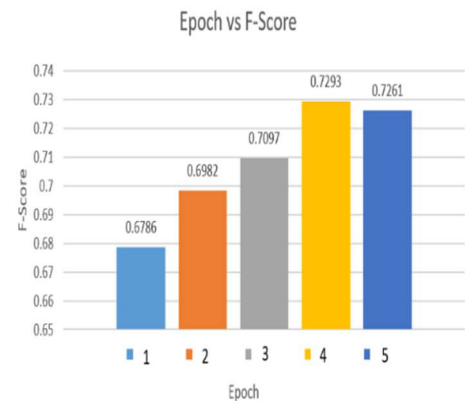


Fig. 5. Results of text detection and recognition of images/ video frames using YOLOv5x

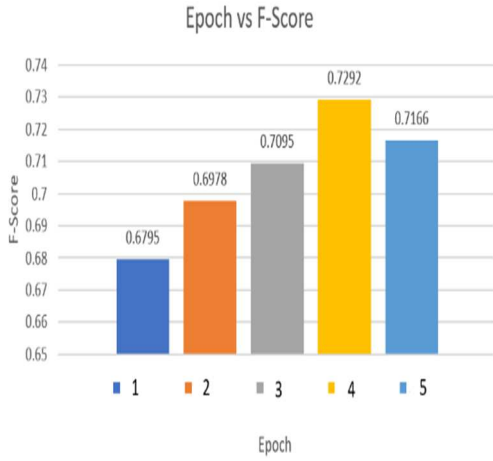
The hyperparameters used in the proposed model are a learning rate of 0.1, batch_size: 16, and the optimizer used SGD-Stochastic Gradient Descent. The middle/hidden layers employ the ReLU-activation function, whereas the final detection layer employs the Sigmoid-activation function. The loss function used is Binary-Cross-Entropy. Moreover, a series of observations were done using different epoch values of 1, 2, 3, and 4, and got a good result at an epoch value of 4 for a smaller set of images, as shown in Fig. 6. Finally, 300 epoch values are set to train larger dataset images.



(a) ICDAR 2013 Dataset



(b) ICDAR 2015 Dataset



(c) YVT dataset

Fig. 6. F-Score results with different epochs for the proposed method using YOLOv5s (a) ICDAR 2013 Dataset, (b) ICDAR 2015 Dataset, and (c) YVT Video Frames Dataset.

The Proposed method works for detecting and recognizing real-time videos. Fig. 7 shows real-time book-cover recognition system results.



Fig. 7. Real-time book-cover recognition system results.

The formulas for calculating Precision, Recall, and F-score is shown in (1), (2), and (3) respectively.

$$\text{Precision} = \frac{\text{Tru.p}}{\text{Tru.p} + \text{Fa.p}} \quad (1)$$

$$\text{Recall} = \frac{\text{Tru.p}}{\text{Tru.p} + \text{Fa.n}} \quad (2)$$

$$\text{F1 - score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

Where True_Positive is Tru.p, False_Positive is Fa.p, and False_Negative is Fa.n. We calculated Tru.p, Fa.p, and Fa.n by comparing the coordinate points of each class to the actual coordinate points (ground-truth). Tables I, II, and III show the text detection and recognition results obtained by ICDAR 2013, ICDAR 2015, and YVT Dataset.

TABLE. I. ICDAR 2013 COMPARISON RESULTS

Method	Precision	Recall	F-score
End-to-end [23]	75.08	52.28	61.64
EAST [24]	64.13	53.22	56.44
End-to-end with optical flow [24]	68.49	55.69	61.43
Proposed Method	62	62	62.00

TABLE. II. ICDAR 2015 COMPARISON RESULTS

Method	Precision	Recall	F-score
StradVision2 [18]	36.74	77.46	49.84
StradVision1 [18]	46.27	53.39	49.57
Proposed Method	61	46	52.44

TABLE. III. YVT COMPARISON RESULTS

Method	Precision	Recall	F-score
Zhao et al. [25]	51.4	52.6	51.99
Shivakumara et al. [26]	57.4	59.8	58.57
Proposed Method	76	55	63.81

Fig. 8 and Fig. 9 show some of the comparison results for text detection between yolov5s and yolov5x. Here, yolov5x can detect text with very high accuracy, whereas yolov5s detect text with less accuracy. The proposed method detected the text with a complex background, and all images are taken from ICDAR 2013, ICDAR 2015, and YVT video frames datasets.

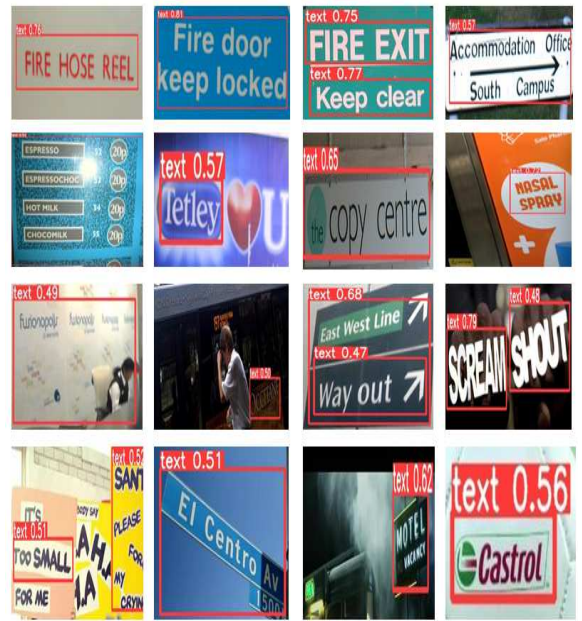


Fig. 8. Examples of scene text detected by the YOLOV5s method.

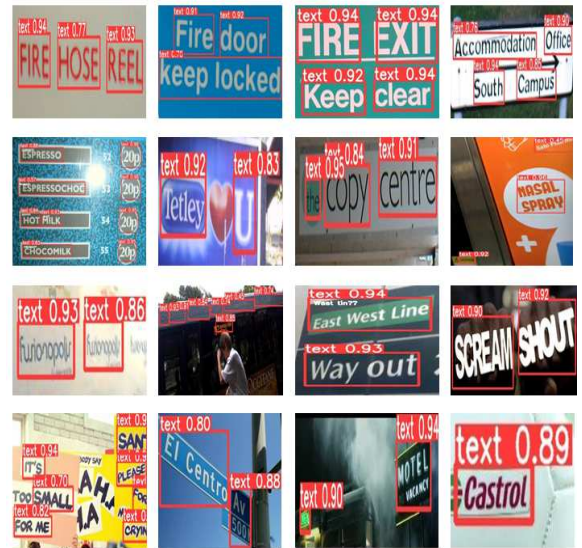


Fig. 9. Examples of scene text detected by the YOLOv5x method.



(a)



(b)

Fig. 10. YOLOv5s results with a non-text bounding box detected as shown in images (a) and (b).

There will be false detection of text in a few cases, as shown in the above Fig. 10(a) and 10(b), and it will not be detected even if the text is present and a bounding box is created for the non-text area. It will be further rectified by fine-tuning with a better confidence value of the YOLOv5x model.

V. CONCLUSION

This work presented a customized YOLOv5x model for text detection and TesseractOCR for text recognition in images/video frames. The customized model helps to detect the text in real-world scenes by fine-tuning the hyperparameter of the YOLOv5x model, and TesseractOCR uses an adaptive thresholding method to effectively extract the character in an image even without character-level annotations. The proposed system is suitable for handling near-horizontal images and complex background scene images. The results show better performance with a good F-score for ICDAR 2013, ICDAR 2015, and YVT video dataset by performing many iterations with different epochs. The proposed method will be used to detect and recognize text in different languages with better performance in future work.

ACKNOWLEDGMENT

We thank the research community and well-wishers.

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